

B6 Condensed-Matter Physics — Model Solutions, 2017

Question 1 — FCC lattice, X-ray diffraction in zincblende GaAs

Problem. Conventional FCC cube of side a . Describe the reciprocal lattice; find d_{111} , the angle between (111) and $(\bar{1}11)$, and between (111) and (100). For zincblende GaAs ($a = 5.653 \times 10^{-10}$ m, basis Ga at 0, As at $\frac{1}{4}\frac{1}{4}\frac{1}{4}$), find scattering angles and relative intensities of the lowest six reflections at $E = 8$ keV. Compare with AlAs, $a = 5.660 \times 10^{-10}$ m.

Reciprocal lattice

Real-space FCC primitive vectors give a BCC reciprocal lattice with conventional cube side $4\pi/a$.

$$\boxed{\text{Reciprocal of FCC}(a) = \text{BCC with conventional cube } 4\pi/a.}$$

The first Brillouin zone is the truncated octahedron (Wigner–Seitz cell of BCC).

Spacing d_{111} and link to reciprocal lattice

Cubic spacing formula:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}.$$

Substitute $(hkl) = (111)$:

$$\boxed{d_{111} = \frac{a}{\sqrt{3}}}.$$

$\mathbf{G}_{111} = (2\pi/a)(1, 1, 1)$ points along the cube body-diagonal; its tip is the L-point on the BZ boundary.

Angle between (111) and $(\bar{1}11)$ planes

Plane normals lie along $\mathbf{G}_{hkl} \propto (h, k, l)$. Angle between normals:

$$\cos \alpha = \frac{(1, 1, 1) \cdot (-1, 1, 1)}{|(1, 1, 1)| |(-1, 1, 1)|} = \frac{1}{3}.$$

$$\boxed{\alpha_{(111),(\bar{1}11)} = \arccos(1/3) \approx 70.53^\circ.}$$

Angle between (111) and (100) planes

$$\cos \beta = \frac{(1, 1, 1) \cdot (1, 0, 0)}{\sqrt{3}} = \frac{1}{\sqrt{3}}.$$

$$\boxed{\beta_{(111),(100)} = \arccos(1/\sqrt{3}) \approx 54.74^\circ.}$$

Powder diffraction from zincblende GaAs

Wavelength from $E = hc/\lambda$:

$$\lambda = \frac{hc}{E} = \frac{(6.626 \times 10^{-34} \text{ J s})(3.0 \times 10^8 \text{ m s}^{-1})}{(8000)(1.602 \times 10^{-19} \text{ J})}.$$

$$\lambda = 1.55 \times 10^{-10} \text{ m} = 1.55 \text{ \AA}.$$

Bragg condition: $\lambda = 2d_{hkl} \sin \theta$, scattering angle 2θ .

Zincblende structure factor (FCC+basis Ga $\mathbf{0}$, As $\frac{1}{4}\frac{1}{4}\frac{1}{4}$):

$$F_{hkl} = F_{\text{FCC}} [f_{\text{Ga}} + f_{\text{As}} e^{i\pi(h+k+l)/2}].$$

$F_{\text{FCC}} = 4$ if h, k, l all even or all odd; zero otherwise.

Three FCC-allowed classes:

- all odd: $|F|^2 \propto f_{\text{Ga}}^2 + f_{\text{As}}^2$ (medium).
- all even, $h + k + l = 4n$: $|F|^2 \propto (f_{\text{Ga}} + f_{\text{As}})^2$ (strong).
- all even, $h + k + l = 4n + 2$: $|F|^2 \propto (f_{\text{Ga}} - f_{\text{As}})^2$ (weak; non-zero because $Z_{\text{Ga}} = 31 \neq Z_{\text{As}} = 33$).

Lowest six reflections by $N = h^2 + k^2 + l^2$, with $\sin \theta = \lambda\sqrt{N}/(2a)$:

$$\frac{\lambda}{2a} = \frac{1.55}{2 \times 5.653} = 0.1371.$$

(hkl)	N	$d_{hkl}/\text{\AA}$	$\sin \theta$	θ	2θ	class
(111)	3	3.264	0.2374	13.74°	27.47°	odd, medium
(200)	4	2.827	0.2741	15.91°	31.82°	even $4n + 2$, weak
(220)	8	1.999	0.3877	22.81°	45.63°	even $4n$, strong
(311)	11	1.704	0.4546	27.04°	54.08°	odd, medium
(222)	12	1.632	0.4748	28.35°	56.70°	even $4n + 2$, weak
(400)	16	1.413	0.5483	33.25°	66.50°	even $4n$, strong

Estimate atomic form factors at $\sin \theta/\lambda$ small using $f \approx Z$:

$$(f_{\text{Ga}} + f_{\text{As}})^2 \approx 64^2 = 4096.$$

$$(f_{\text{Ga}}^2 + f_{\text{As}}^2) \approx 31^2 + 33^2 = 2050.$$

$$(f_{\text{As}} - f_{\text{Ga}})^2 \approx 2^2 = 4.$$

Multiplicities m_{hkl} : (111):8, (200):6, (220):12, (311):24, (222):8, (400):6.

Relative intensity $I \propto m_{hkl}|F_{hkl}|^2$ (Lorentz–polarisation/Debye–Waller suppressed):

$$\begin{aligned} I_{111} : I_{200} : I_{220} : I_{311} : I_{222} : I_{400} \\ \propto 8(2050) : 6(4) : 12(4096) : 24(2050) : 8(4) : 6(4096) \\ = 16400 : 24 : 49152 : 49200 : 32 : 24576. \end{aligned}$$

Normalise to (220) = 100:

$$I_{111}:I_{200}:I_{220}:I_{311}:I_{222}:I_{400} \approx 33:0.05:100:100:0.07:50.$$

The (200) and (222) peaks are nearly extinct: they would vanish exactly if Ga and As were identical (as in the diamond structure of Si).

AlAs comparison: $\Delta(2\theta)$ for the (400) reflection

Differentiate Bragg's law at fixed λ :

$$\cos \theta \, d\theta = -\frac{\lambda\sqrt{N}}{2} \frac{da}{a^2} = -\sin \theta \frac{da}{a}.$$

$$d(2\theta) = -2 \tan \theta \frac{da}{a}.$$

For (400) with $\theta = 33.25^\circ$ and $\Delta a/a = 7 \times 10^{-4}/5.653 = 1.24 \times 10^{-3}$:

$$\Delta(2\theta) = -2 \tan(33.25^\circ)(1.24 \times 10^{-3}) = -1.62 \times 10^{-3} \text{ rad}.$$

$\Delta(2\theta)_{(400)} \approx -0.093^\circ \quad (\text{AlAs at smaller } 2\theta).$

This is at the edge of a standard powder diffractometer ($\sim 0.05^\circ$); higher-order reflections amplify $\tan \theta$ and so the splitting. Alternatively, exploit the Z -difference: $Z_{\text{Al}} = 13$ vs $Z_{\text{Ga}} = 31$ change $|F|^2$ for the all-odd class by a factor $(13^2 + 33^2)/(31^2 + 33^2) \approx 0.62$, and the weak even “ $4n + 2$ ” reflections (e.g. (200), (222)) become much stronger in AlAs because $|f_{\text{As}} - f_{\text{Al}}| = 20$ vs 2 for GaAs. *Resonant (anomalous) scattering* near a Ga or As K-edge is the cleanest discriminator.

Question 2 — Free-electron metal: heat capacity, entropy, Pauli χ , Na

Problem. From $U = \int_0^\infty E g(E) f(E) dE$ explain g , f and their T -dependence; argue $C = \gamma T$ with $\gamma = \alpha k_B^2 g(E_F)$. Derive low- T entropy and Pauli susceptibility. For Na (BCC, monovalent, $R_H = -25.0 \times 10^{-11} \text{ m}^3 \text{ C}^{-1}$, $m^* = 1.3m_e$), find a , E_F , γ , χ_p .

Meaning of $g(E)$ and $f(E)$; rough $C = \gamma T$ argument

$g(E) dE$: number of single-electron states per unit volume in $[E, E + dE]$ (sums spin). Independent of T .

$f(E) = [\exp((E - \mu)/k_B T) + 1]^{-1}$: Fermi–Dirac occupation. Carries all T -dependence (also weakly through $\mu(T)$).

Only electrons within $\sim k_B T$ of E_F are thermally active.

$$\Delta N \sim g(E_F) k_B T.$$

Each gains energy $\sim k_B T$:

$$\Delta U \sim g(E_F) (k_B T)^2.$$

Differentiate w.r.t. T :

$$C = \frac{\partial U}{\partial T} \sim 2k_B^2 g(E_F) T.$$

Exact Sommerfeld coefficient is $\alpha = \pi^2/3$:

$C = \gamma T, \quad \gamma = \frac{\pi^2}{3} k_B^2 g(E_F).$

Low-temperature entropy

Use $C = T(\partial S/\partial T)_V$:

$$\frac{\partial S}{\partial T} = \frac{C}{T} = \gamma.$$

Integrate from $T = 0$ (third law: $S(0) = 0$):

$S(T) = \gamma T.$

Pauli susceptibility

Apply $\mathbf{B} = B\hat{\mathbf{z}}$. Spin Zeeman energy $\mp\mu_B B$ shifts the spin-up/down half-bands:

$$g_{\uparrow}(E) = \frac{1}{2}g(E + \mu_B B), \quad g_{\downarrow}(E) = \frac{1}{2}g(E - \mu_B B).$$

Net magnetisation per volume:

$$M = \mu_B \int_0^\infty [g_{\uparrow}(E) - g_{\downarrow}(E)] f(E) \, dE.$$

Linearise in B via $g(E \pm \mu_B B) \approx g(E) \pm \mu_B B g'(E)$:

$$M = \mu_B^2 B \int_0^\infty [-g'(E)] f(E) \, dE.$$

Integrate by parts; at $T \rightarrow 0$, $-\partial f/\partial E = \delta(E - E_F)$:

$$M = \mu_B^2 g(E_F) B.$$

Susceptibility $\chi_p = \mu_0 M/B$:

$\chi_p = \mu_0 \mu_B^2 g(E_F).$

Sodium: lattice constant from R_H

Free-electron Hall coefficient:

$$R_H = -\frac{1}{ne}.$$

Solve for n :

$$n = -\frac{1}{R_H e} = \frac{1}{(25.0 \times 10^{-11})(1.602 \times 10^{-19})} = 2.50 \times 10^{28} \text{ m}^{-3}.$$

BCC monovalent: 2 atoms per conventional cell, each donates one electron, so $n = 2/a^3$:

$$a = \left(\frac{2}{n}\right)^{1/3} = \left(\frac{2}{2.50 \times 10^{28}}\right)^{1/3}.$$

$a_{\text{Na}} = 4.31 \times 10^{-10} \text{ m} = 4.31 \text{ \AA}.$

Sodium: E_F , γ , χ_p **with** $m^* = 1.3m_e$

Fermi wave-vector:

$$k_F = (3\pi^2 n)^{1/3} = (3\pi^2 \times 2.50 \times 10^{28})^{1/3} = 9.06 \times 10^9 \text{ m}^{-1}.$$

Fermi energy:

$$E_F = \frac{\hbar^2 k_F^2}{2m^*} = \frac{(1.055 \times 10^{-34})^2 (9.06 \times 10^9)^2}{2(1.3)(9.11 \times 10^{-31})}.$$

$$E_F = 3.84 \times 10^{-19} \text{ J}.$$

$$E_F \approx 2.40 \text{ eV}.$$

Density of states at E_F (free electron, per volume):

$$g(E_F) = \frac{3n}{2E_F} = \frac{3(2.50 \times 10^{28})}{2(3.84 \times 10^{-19})} = 9.76 \times 10^{46} \text{ J}^{-1} \text{ m}^{-3}.$$

Electronic specific-heat coefficient (??):

$$\gamma_V = \frac{\pi^2}{3} k_B^2 g(E_F) = \frac{\pi^2}{3} (1.381 \times 10^{-23})^2 (9.76 \times 10^{46}).$$

$$\gamma_V = 61.2 \text{ J K}^{-2} \text{ m}^{-3}.$$

Convert to per mole using molar volume $V_m = N_A/n$:

$$\gamma = \gamma_V \frac{N_A}{n} = (61.2) \frac{6.022 \times 10^{23}}{2.50 \times 10^{28}}.$$

$$\gamma_{\text{Na}} \approx 1.48 \text{ mJ mol}^{-1} \text{ K}^{-2}.$$

(Experimental: $\sim 1.4 \text{ mJ mol}^{-1} \text{ K}^{-2}$ – consistent.)

Pauli susceptibility (??):

$$\chi_p = \mu_0 \mu_B^2 g(E_F) = (4\pi \times 10^{-7}) (9.274 \times 10^{-24})^2 (9.76 \times 10^{46}).$$

$$\chi_p \approx 1.05 \times 10^{-5} \quad (\text{SI, dimensionless}).$$

Question 3 — Curie law, Curie–Weiss, mean-field ferromagnetism in Ni

Problem. Spin-1/2 paramagnet, n moments per volume in field B at T : derive M and $\chi = C/T$ with $C = n\mu_0\mu_B^2/k_B$. Add mean-field $B_{\text{eff}} = \lambda\mu_0 M$: derive Curie–Weiss form $\chi = C/(T - \theta)$. For Ni ($\rho = 8908 \text{ kg m}^{-3}$, $M_r = 58.693$, $T_C = 627 \text{ K}$), find λ and B_{eff} .

(a) Curie law, spin-1/2 paramagnet

Two states $m_s = \pm 1/2$, energies $\mp \mu_B B$. Boltzmann weights $e^{\pm \mu_B B / k_B T}$:

$$\langle m_z \rangle = \mu_B \frac{e^{\mu_B B / k_B T} - e^{-\mu_B B / k_B T}}{e^{\mu_B B / k_B T} + e^{-\mu_B B / k_B T}} = \mu_B \tanh\left(\frac{\mu_B B}{k_B T}\right).$$

Magnetisation per volume:

$$M = n \mu_B \tanh\left(\frac{\mu_B B}{k_B T}\right).$$

Linear regime $\mu_B B \ll k_B T$, $\tanh x \approx x$:

$$M = \frac{n \mu_B^2 B}{k_B T}.$$

Susceptibility $\chi = \mu_0 M / B$:

$$\chi = \frac{C}{T}, \quad C = \frac{n \mu_0 \mu_B^2}{k_B}.$$

(b) Mean-field Curie–Weiss

Replace external field by $B_{\text{tot}} = B + \lambda \mu_0 M$ in the linearised result:

$$M = \frac{C}{\mu_0 T} (B + \lambda \mu_0 M).$$

Collect M :

$$M \left(1 - \frac{\lambda C}{T}\right) = \frac{CB}{\mu_0 T}.$$

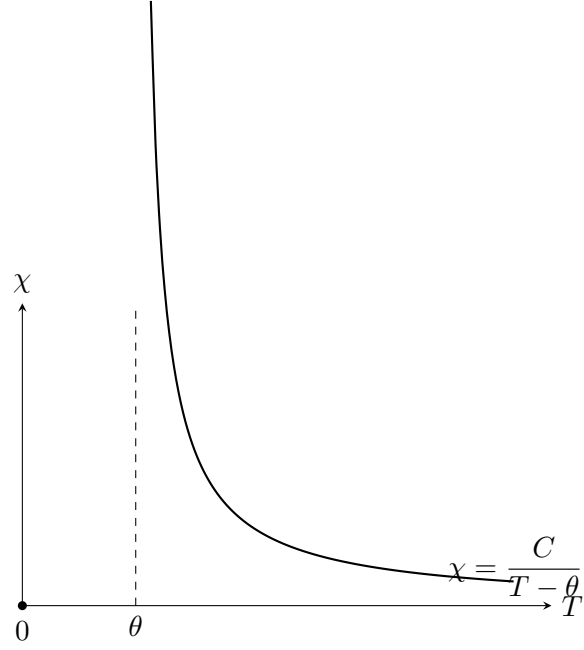
Solve for M :

$$M = \frac{CB}{\mu_0 (T - \lambda C)}.$$

Susceptibility:

$$\chi = \frac{C}{T - \theta}, \quad \theta = \lambda C = \frac{\lambda n \mu_0 \mu_B^2}{k_B}.$$

Significance: θ is the mean-field critical temperature (T_C). $\theta > 0$ favours ferromagnetic alignment; $\theta < 0$ antiferromagnetic. χ diverges as $T \rightarrow \theta^+$, signalling spontaneous order.



(c) Nickel: mean-field λ and B_{eff}

Number density of Ni atoms:

$$n = \frac{\rho N_A}{M_r} = \frac{(8908)(6.022 \times 10^{23})}{58.693 \times 10^{-3}} = 9.14 \times 10^{28} \text{ m}^{-3}.$$

Curie constant from (??):

$$C = \frac{n \mu_0 \mu_B^2}{k_B} = \frac{(9.14 \times 10^{28})(4\pi \times 10^{-7})(9.274 \times 10^{-24})^2}{1.381 \times 10^{-23}}.$$

$$C = 0.715 \text{ K}.$$

Identify $\theta = T_C$ and use (??):

$$\lambda = \frac{T_C}{C} = \frac{627}{0.715}.$$

$$\boxed{\lambda \approx 8.8 \times 10^2.}$$

Saturation magnetisation (spin-1/2, all moments aligned, low T):

$$M_{\text{sat}} = n \mu_B = (9.14 \times 10^{28})(9.274 \times 10^{-24}) = 8.48 \times 10^5 \text{ A m}^{-1}.$$

Mean-field exchange field:

$$B_{\text{eff}} = \lambda \mu_0 M_{\text{sat}} = (878)(4\pi \times 10^{-7})(8.48 \times 10^5).$$

$$\boxed{B_{\text{eff}} \approx 9.3 \times 10^2 \text{ T}.$$

This is two orders of magnitude beyond the strongest laboratory magnets and far exceeds dipolar fields ($\mu_0 \mu_B / r^3 \sim 0.1 \text{ T}$). B_{eff} cannot be a real magnetic field; it parametrises the *quantum exchange interaction*, which is electrostatic (Coulomb + Pauli) in origin.

(d) Why room-temperature Ni often shows $M = 0$

Below T_C a bulk specimen breaks into magnetic *domains* of opposite magnetisation, separated by Bloch walls. Domain formation lowers magnetostatic (stray-field) energy; the macroscopic vector sum of domain magnetisations averages to zero in an unmagnetised sample, even though each domain is locally saturated.

Question 4 — Intrinsic and doped semiconductors

Problem. Show $np = A(m_e^*m_h^*)^B T^C \exp(-E_g/k_B T)$, identify A, B, C . Find $\mu(T)$ for intrinsic case and the constant α . For Si ($E_g = 1.11$ eV, $m_e^* = 0.26m_e$, $m_h^* = 0.39m_e$, $\epsilon_r = 11.7$): donor binding energy of P; p at $T = 300$ K for pure and $N_d = 10^{20} \text{ m}^{-3}$ doped Si.

(a) Carrier product np

Conduction-band density of states (parabolic, from E_c):

$$g_c(E) = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{3/2} (E - E_c)^{1/2}.$$

Boltzmann tail $f(E) \approx e^{-(E-\mu)/k_B T}$ valid when $E_c - \mu \gg k_B T$. Electron number density:

$$n = \int_{E_c}^{\infty} g_c(E) e^{-(E-\mu)/k_B T} dE.$$

Substitute $x = (E - E_c)/k_B T$:

$$n = \frac{1}{2\pi^2} \left(\frac{2m_e^* k_B T}{\hbar^2} \right)^{3/2} e^{-(E_c-\mu)/k_B T} \int_0^{\infty} x^{1/2} e^{-x} dx.$$

Use $\int_0^{\infty} x^{1/2} e^{-x} dx = \sqrt{\pi}/2$:

$$n = 2 \left(\frac{m_e^* k_B T}{2\pi \hbar^2} \right)^{3/2} e^{-(E_c-\mu)/k_B T} \equiv N_c e^{-(E_c-\mu)/k_B T}.$$

Holes from $1 - f(E) \approx e^{-(\mu-E)/k_B T}$ in valence band, mirrored:

$$p = 2 \left(\frac{m_h^* k_B T}{2\pi \hbar^2} \right)^{3/2} e^{-(\mu-E_v)/k_B T} \equiv N_v e^{-(\mu-E_v)/k_B T}.$$

Multiply (??) and (??); μ cancels:

$$np = 4 \left(\frac{k_B T}{2\pi \hbar^2} \right)^3 (m_e^* m_h^*)^{3/2} e^{-E_g/k_B T}, \quad E_g = E_c - E_v.$$

$$np = A (m_e^* m_h^*)^B T^C e^{-E_g/k_B T}, \quad A = \frac{4k_B^3}{(2\pi \hbar^2)^3}, \quad B = \frac{3}{2}, \quad C = 3.$$

(b) Intrinsic chemical potential $\mu(T)$

Charge neutrality (intrinsic): $n = p$. Equate (??) and (??):

$$N_c e^{-(E_c - \mu)/k_B T} = N_v e^{-(\mu - E_v)/k_B T}.$$

Take logarithm:

$$-\frac{E_c - \mu}{k_B T} + \ln N_c = -\frac{\mu - E_v}{k_B T} + \ln N_v.$$

Solve for μ :

$$2\mu = E_c + E_v + k_B T \ln(N_v/N_c).$$

Substitute $N_v/N_c = (m_h^*/m_e^*)^{3/2}$:

$$\mu(T) = \frac{E_c + E_v}{2} + \frac{3}{4} k_B T \ln \frac{m_h^*}{m_e^*}, \quad \alpha = \frac{3}{4} \ln(m_h^*/m_e^*).$$

For Si: $\alpha = \frac{3}{4} \ln(0.39/0.26) = 0.304$.

(c) Phosphorus donor binding energy

Hydrogenic donor in dielectric medium with effective mass m_e^* :

$$E_R^* = R_H \frac{m_e^*/m_e}{\epsilon_r^2} = (13.6 \text{ eV}) \frac{0.26}{11.7^2}.$$

$$E_R^* \approx 26 \text{ meV}.$$

At $T = 1000 \text{ K}$, $k_B T \approx 86 \text{ meV} \gg E_R^*$: donors are fully ionised (in fact already so above $\sim 50 \text{ K}$).

(d) Hole density at $T = 300 \text{ K}$: pure and P-doped Si

Effective densities of states at $T = 300 \text{ K}$:

$$N_c = 2 \left(\frac{(0.26)(9.109 \times 10^{-31})(1.381 \times 10^{-23})(300)}{2\pi(1.055 \times 10^{-34})^2} \right)^{3/2}.$$

$$N_c = 3.33 \times 10^{24} \text{ m}^{-3}.$$

$$N_v = N_c \left(\frac{0.39}{0.26} \right)^{3/2} = 6.11 \times 10^{24} \text{ m}^{-3}.$$

Intrinsic carrier density:

$$n_i^2 = N_c N_v e^{-E_g/k_B T}.$$

Evaluate exponent: $E_g/k_B T = 1.11/(2.585 \times 10^{-2}) = 42.94$:

$$e^{-42.94} = 2.25 \times 10^{-19}.$$

$$n_i^2 = (3.33 \times 10^{24})(6.11 \times 10^{24})(2.25 \times 10^{-19}) = 4.58 \times 10^{30} \text{ m}^{-6}.$$

$$n_i = 2.14 \times 10^{15} \text{ m}^{-3}.$$

(i) Pure Si: $n = p = n_i$:

$$p_{\text{intr}} \approx 2.1 \times 10^{15} \text{ m}^{-3}.$$

μ sits near mid-gap; the small shift $\alpha k_B T = (0.304)(0.0259 \text{ eV}) = 7.9 \text{ meV}$ above mid-gap (because $m_h^* > m_e^*$).

(ii) n -type, $N_d = 10^{20} \text{ m}^{-3}$ (fully ionised, $N_d \gg n_i$): $n \approx N_d$ and $p = n_i^2/n$:

$$p = \frac{n_i^2}{N_d} = \frac{4.58 \times 10^{30}}{10^{20}}.$$

$$p_{n\text{-type}} \approx 4.6 \times 10^{10} \text{ m}^{-3}.$$

Position of μ : $\mu = E_c - k_B T \ln(N_c/N_d)$:

$$\mu = E_c - (0.0259 \text{ eV}) \ln(3.33 \times 10^{24}/10^{20}) = E_c - 0.27 \text{ eV}.$$

μ has shifted from near mid-gap up to $\sim 0.27 \text{ eV}$ below E_c – well into the upper half of the gap, as required for an n -type material.